

Food Remedy API

Research Report – RE013

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RE013 – Consumer Behavior and Food Choices Influenced by Hormones

Goal: Understand user behavior to support personalized and adaptive nutrition systems.

1. Introduction

Food choices are influenced not only by personal preference and environment but also by biological and hormonal processes. Hormones such as ghrelin, leptin, cortisol, estrogen, and progesterone play a key role in regulating hunger, cravings, mood, and eating behavior.

These hormonal signals interact with the brain's reward system, which explains why users often crave specific foods (e.g., sugary or high-fat foods) during stress or certain physiological phases.

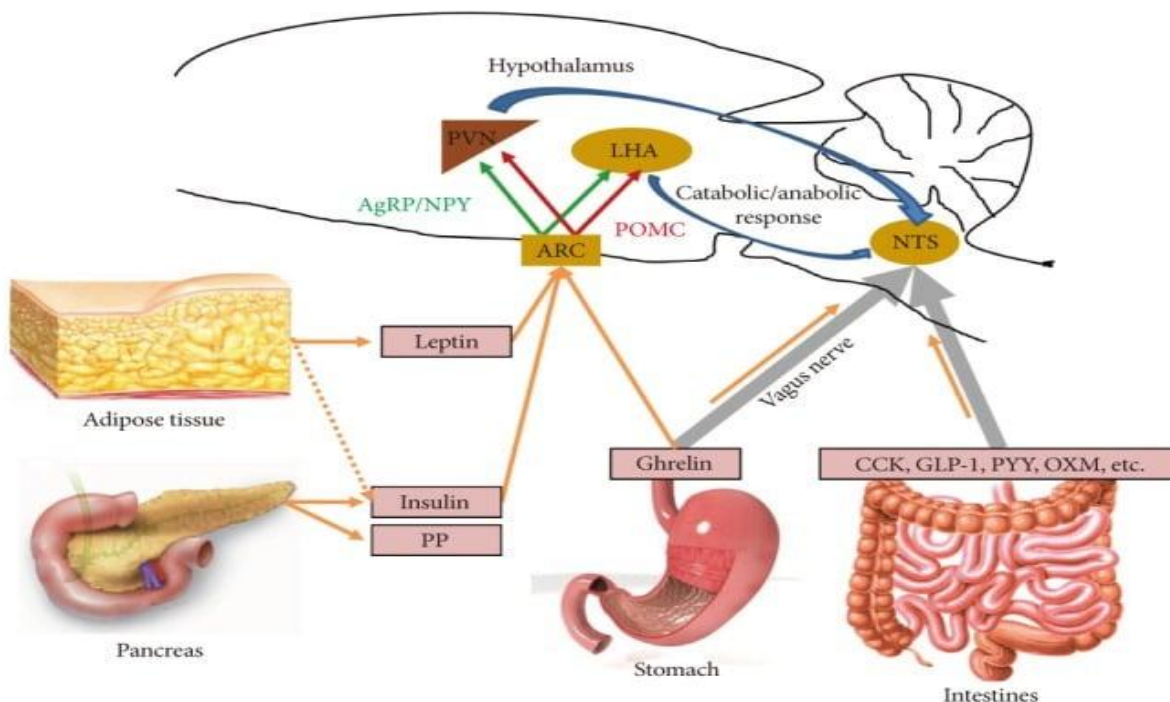
For Food Remedy, understanding these behaviors is important to design a personalized nutrition system that adapts to users' changing needs rather than providing static recommendations.

2. Hormonal Influence on Hunger, Cravings, and Food Preferences

Hormones regulate appetite through complex communication between the gut, brain, and endocrine system.

- Ghrelin increases hunger and stimulates food intake
- Leptin signals satiety and reduces appetite
- Cortisol increases appetite during stress
- Estrogen tends to suppress appetite
- Progesterone may increase cravings and food intake

Research shows that hormonal fluctuations can influence not only how much users eat but also what types of foods they prefer, often increasing desire for energy-dense foods such as sweets and fast food



In individuals with menstrual cycles, appetite and cravings often increase during the luteal (premenstrual) phase, which is linked to hormonal shifts involving estrogen and progesterone. This explains recurring patterns of increased hunger or emotional eating during this time.

3. Behavioural Patterns: Emotional Eating and Stress Eating

Hormonal effects strongly influence behavioral eating patterns.

Emotional Eating

Emotional eating occurs when individuals eat in response to feelings such as stress, sadness, or boredom rather than physical hunger. Hormonal changes can intensify these behaviors by affecting mood and reward sensitivity.

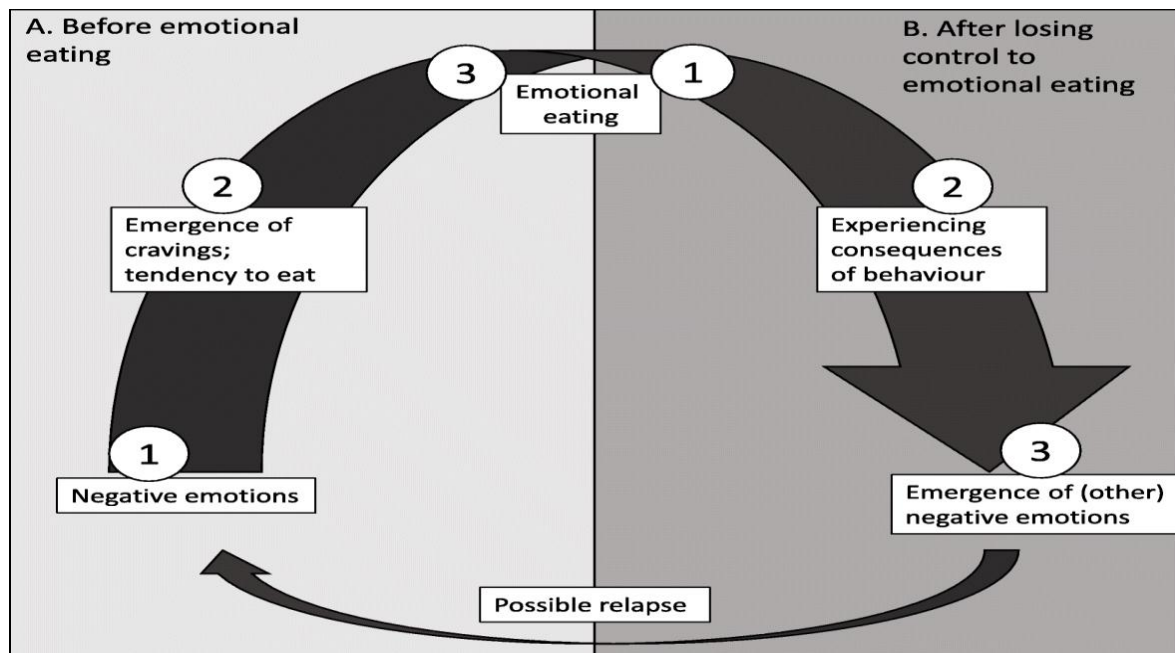
Stress Eating

Stress activates the release of cortisol, which increases appetite and promotes cravings for high-fat and high-sugar foods. This leads to **reward-driven eating**, where users prefer comfort foods over healthier options.

Cycle-Based Behaviour

Hormonal fluctuations across menstrual cycles can increase vulnerability to:

- Cravings
- Snacking frequency
- Binge-like eating patterns



These behaviours show that food decisions are often influenced by **internal biological states**, not just conscious choice.

4. Impact of Awareness on Food Choices

Awareness of hormonal effects can significantly improve user behaviour.

When users understand that cravings or hunger changes are hormonally influenced, they are more likely to:

- **Anticipate changes** (e.g., prepare meals in advance)
- **Reduce guilt** associated with cravings
- **Make informed decisions** instead of impulsive choices

For example, a user aware of premenstrual cravings may choose healthier alternatives rather than high-sugar snacks.

Studies show that knowledge of hormonal and menstrual health improves **self-management and decision-making**, leading to better long-term outcomes.

5. Role of Nutrition Systems and Apps

Digital nutrition systems can play a key role in supporting users during hormonal changes.

Modern apps such as Clue and Flo demonstrate how tracking and personalized insights can help users understand their bodies.

Key features that support users include:

1. Tracking and Pattern Recognition

- Track hunger, cravings, mood, and cycle phase
- Identify behavioural patterns over time

2. Personalised Insights

- Provide feedback like:
“You tend to crave sweets during this phase”

3. Adaptive Recommendations

- Suggest food options based on current state
- Move beyond generic diet advice

4. Behavioural Support

- Stress management suggestions
- Reminders for hydration and sleep

6. Evaluation

Hormonal influence on food behaviour is well supported by research; however:

- Effects vary between individuals
- Not all users experience strong fluctuations
- Some research areas (e.g., menstrual diet patterns) still require more evidence

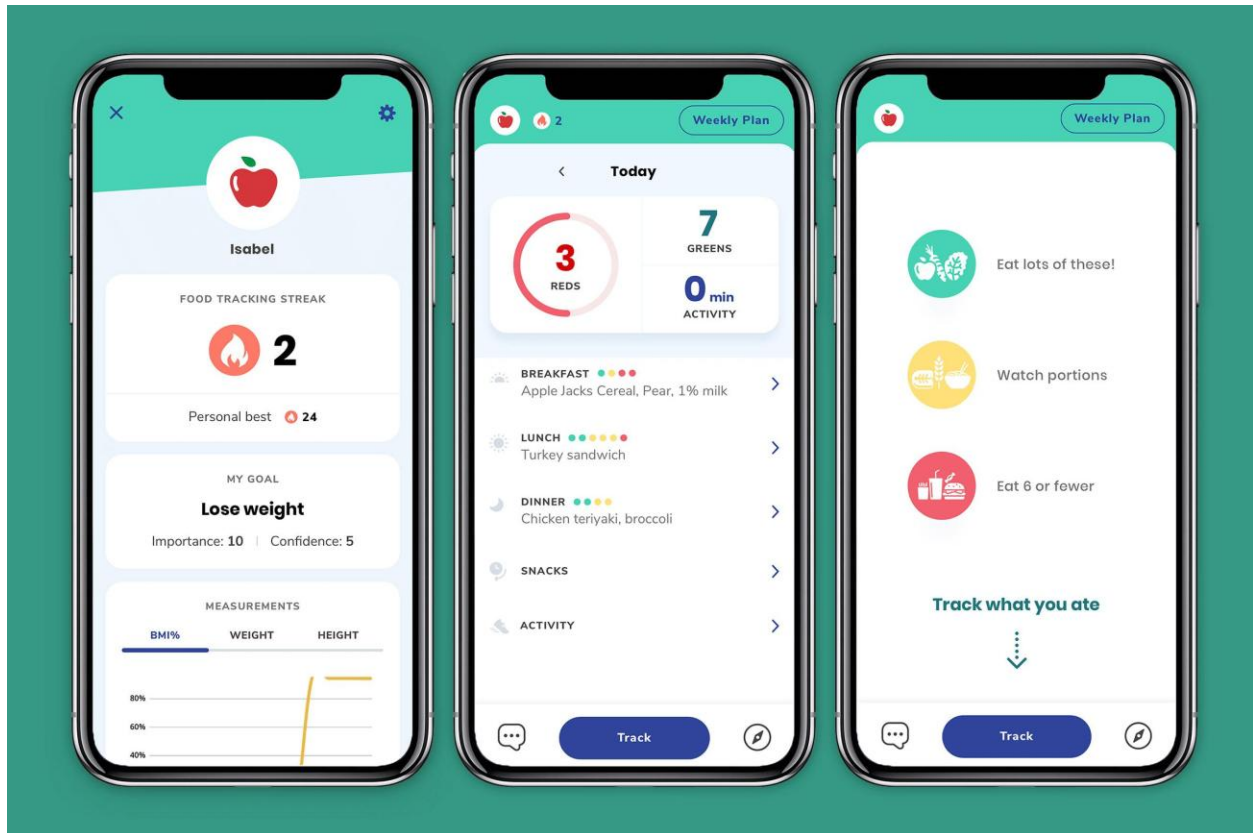
Therefore, nutrition systems should focus on **personalization rather than generalization**.

Instead of assuming behaviour, systems should learn from user data and adjust recommendations accordingly.

7. Recommendations for Food Remedy

Based on the analysis, Food Remedy should:

- Implement **craving, mood, and hunger tracking**
- Use **pattern detection** to identify behavioural trends
- Provide **hormone-aware insights** in simple language
- Offer **adaptive food recommendations**
- Support users during **high-risk periods (stress or hormonal changes)**
- Include **non-food coping strategies** (e.g., stress management)
- Ensure **privacy and secure handling of user data**



8. Conclusion

Hormonal fluctuations significantly influence hunger, cravings, and food behaviour. These biological changes interact with emotional and environmental factors, leading to patterns such as stress eating and emotional eating.

By integrating hormone-aware insights and personalized recommendations, Food Remedy can create a system that is **adaptive, user-centered, and behaviourally effective**, improving both user experience and health outcomes.

9. References

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